Emmanuel Muwanguzi

London, UK | +447412347256 | muwanguziemmah64@gmail.com | emmanuelmuwanguzi-portfolio.netlify.app

SUMMARY

Master’s graduate with hands-on research and development experience in machine learning, computer vision, and web based artificial intelligence applications. Completed a research project titled “Explainable Web-Based Tomato Disease Detection Using Lightweight CNN”, involving the development and local deployment of a lightweight convolutional neural network for plant disease detection. Experienced in Python, machine learning model development, Flask backend development, and React frontend development. Interested in Research Assistant opportunities involving machine learning, computer vision, artificial intelligence, image classification, agricultural technology, and applied AI research.

EDUCATION

Roehampton University, London *MSc Data Science, Master’s Dissertation: Web-Based Plant Disease Detection Using Lightweight CNN,*

[2025 – 2026]

Relevant areas: Machine Learning, Artificial Intelligence, Computer Vision, Deep Learning, Data Analysis

Makerere University Kampala *BSc Computer Science, Relevant areas:, Data and Information Management, Systems Analysis and Design, Emerging Trends in Computer Science, Artificial Intelligence [2019 – 2023]*

EXPERIENCE

Master’s Research Project — Explainable Web-Based Tomato Disease Detection Using Lightweight CNN

Roehampton University
2026

- Investigated the application of lightweight Convolutional Neural Networks (CNNs) for automated plant disease detection from plant leaf images.
- Developed and evaluated a machine learning model for image-based plant disease classification using Python.
- Performed data preparation and preprocessing required for training and evaluating the plant disease detection model.
- Designed a web-based application to provide an interface for interacting with the trained machine learning model.
- Developed the application frontend using React to enable users to submit plant images and receive model predictions.
- Developed a Flask-based backend to connect the web interface with the machine learning model.
- Integrated the trained CNN model into the web application for local deployment and inference.
- Tested the complete application workflow, including image input, model inference, backend processing, and presentation of prediction results through the frontend.
- Analysed model performance and considered the practical requirements and limitations of deploying lightweight deep-learning models in an application environment.
- Documented the research methodology, implementation, experiments, results, and limitations in a Master’s dissertation.
- Technologies: Python, Convolutional Neural Networks (CNN), Machine Learning, Computer Vision, React, Flask, TensorFlow/PyTorch
- Research Focus: Plant Disease Detection | Image Classification | Lightweight Deep Learning | Computer Vision | Applied Artificial Intelligence

PROJECTS

Extracting Urban Infrastructure from Satellite Imagery | Roehampton University | 2026

- Developed an image-processing and computer vision approach for extracting urban infrastructure features from satellite imagery.
- Pre-processed and analysed satellite images to identity and distinguish urban infrastructure from surrounding land-cover areas.
- Applied images processing, feature extraction and classification techniques to identify relevant infrastructure from satellite imagery.
- Investigated challenges associated with satellite image analysis including variations in image characteristics, complex urban environments, and similarities between different land-cover features.
- Technologies: MATLAB

Image Captioning | Roehampton University | 2026

- Developed an image captioning system to automatically generate natural-language descriptions of input images.
- Implemented a deep learning-based image-to-text approach combining a convolutional neural network (CNN) for visual feature extraction.
- Prepared and pre-processed image-caption data, trained the model, and evaluated its ability to generate captions for previously unseen images.
- Technologies: Python, TensorFlow/PyTorch, CNN, Natural Language Processing (NLP), Deep Learning, Transformers, NumPy, Pandas, OpenCV

TECHNICAL SKILLS

Python
JavaScript
Convolutional Neural Networks (CNN)
Deep Learning
Image Classification
Machine Learning Model Development
Model Evaluation
Data Preprocessing
TensorFlow/PyTorch
Scikit-learn
OpenCV
Research Problem Formulation
Literature Review
Experimental Design
Data Collection and Preparation
Model Training and Evaluation
Quantitative Analysis
Results Interpretation
Technical Documentation
Scientific/Academic Writing
React
Flask
REST APIs
Frontend and Backend Integration
Local Machine Learning Model Deployment

Git/GitHub
Jupyter Notebook
VS Code

RESEARCH INTERESTS

- Machine Learning - Deep Learning - Computer Vision - Image Classification - Agricultural Artificial Intelligence - Plant Disease Detection - Lightweight Neural Networks - Applied Artificial Intelligence - Machine Learning Model Deployment

REFERENCES

Available upon request.